# SG No. 75 $C_4^1$ $P4$ [ tetragonal ]

* plus set: $+[0, 0, 0]$

* Wyckoff site: `1a`, site symmetry: `4..`

Table 1: Wyckoff bond: `1a@1a`, bond symmetry: `4..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $[0, 0, z]$ | `[1,2,3,4]` |

Table 2: Wyckoff bond: `2b@1a`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $[0, 0, z]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $[0, 0, z]$ | `[3,-4]` |

Table 3: Wyckoff bond: `4c@1a`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[0, 0, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[0, 0, z]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $[0, 0, z]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $[0, 0, z]$ | `[4]` |

* Wyckoff site: `1b`, site symmetry: `4..`

Table 4: Wyckoff bond: `1a@1b`, bond symmetry: `4..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[1,2,3,4]` |

Table 5: Wyckoff bond: `2b@1b`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[3,-4]` |

Table 6: Wyckoff bond: `4c@1b`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | `[4]` |

* Wyckoff site: `2c`, site symmetry: `2..`

Table 7: Wyckoff bond: `2a@2c`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, 0]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,-2]` |
| 2 | $[-Y, X, 0]$ | $\left[\frac{1}{2}, 0, z\right]$ | `[3,-4]` |

Table 8: Wyckoff bond: `2b@2c`, bond symmetry: `2..`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[0, 0, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1,2]` |
| 2 | $[0, 0, Z]$ | $\left[\frac{1}{2}, 0, z\right]$ | `[3,4]` |

Table 9: Wyckoff bond: `4c@2c`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $\left[0, \frac{1}{2}, z\right]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $\left[\frac{1}{2}, 0, z\right]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $\left[\frac{1}{2}, 0, z\right]$ | `[4]` |

* Wyckoff site: `4d`, site symmetry: `1`

Table 10: Wyckoff bond: `4a@4d`, bond symmetry: `1`

| No. | vector | center | mapping |
|---|---|---|---|
| 1 | $[X, Y, Z]$ | $[x, y, z]$ | `[1]` |
| 2 | $[-X, -Y, Z]$ | $[-x, -y, z]$ | `[2]` |
| 3 | $[-Y, X, Z]$ | $[-y, x, z]$ | `[3]` |
| 4 | $[Y, -X, Z]$ | $[y, -x, z]$ | `[4]` |